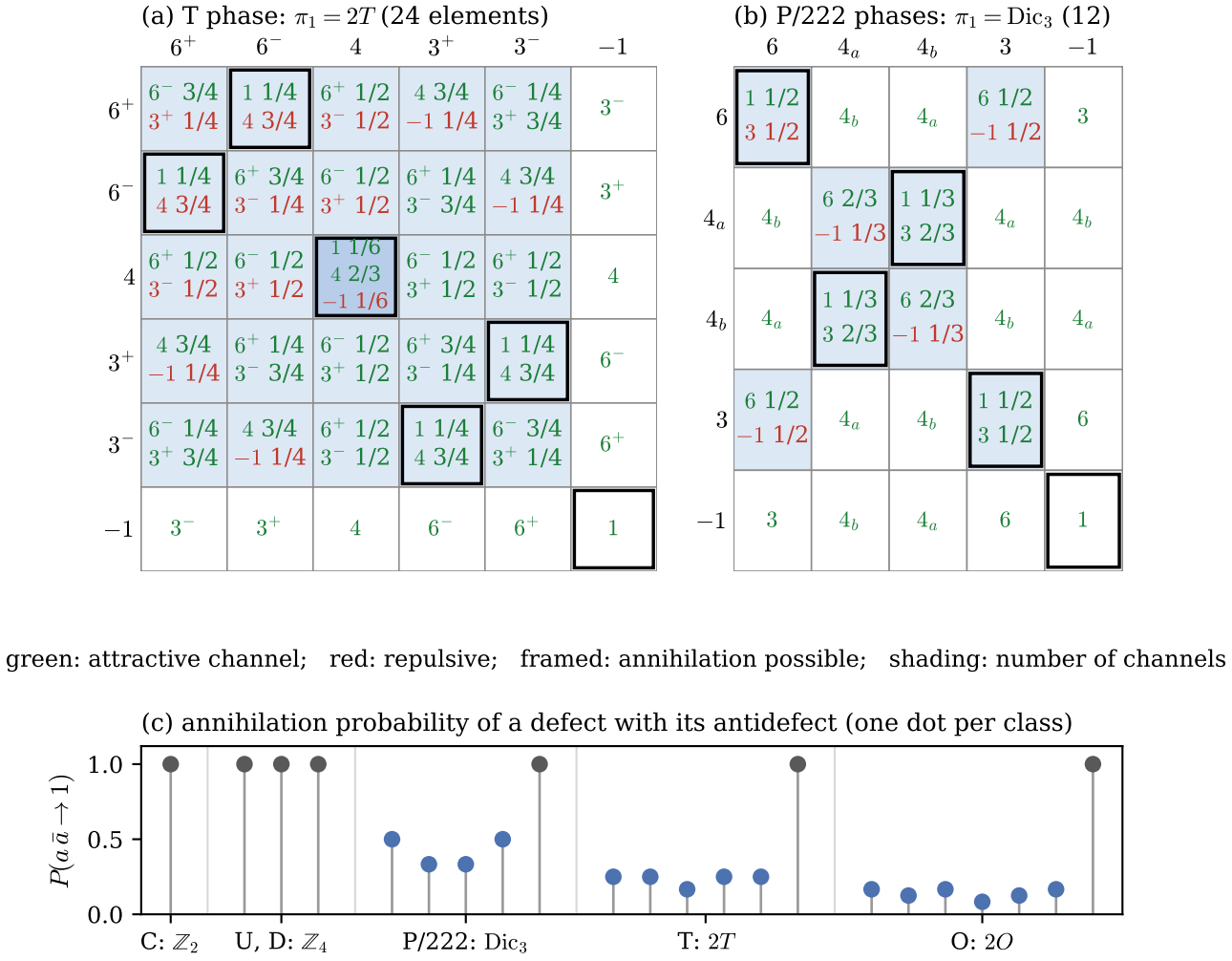
(a) T phase: $\pi_1 = 2T$ (24 elements)
6^+ 6^- 4 3^+ 3^- -1
6^+: $6^- \ 3/4$, $3^+ \ 1/4$ | $1 \ 1/4$, $4 \ 3/4$ | $6^+ \ 1/2$, $3^- \ 1/2$ | $4 \ 3/4$, $-1 \ 1/4$ | $6^- \ 1/4$, $3^+ \ 3/4$ | 3^-
6^-: $1 \ 1/4$, $4 \ 3/4$ | $6^+ \ 3/4$, $3^- \ 1/4$ | $6^- \ 1/2$, $3^+ \ 1/2$ | $6^+ \ 1/4$, $3^- \ 3/4$ | $4 \ 3/4$, $-1 \ 1/4$ | 3^+
4: $6^+ \ 1/2$, $3^- \ 1/2$ | $6^- \ 1/2$, $3^+ \ 1/2$ | $1 \ 1/6$, $4 \ 2/3$, $-1 \ 1/6$ | $6^- \ 1/2$, $3^+ \ 1/2$ | $6^+ \ 1/2$, $3^- \ 1/2$ | 4
3^+: $4 \ 3/4$, $-1 \ 1/4$ | $6^+ \ 1/4$, $3^- \ 3/4$ | $6^- \ 1/2$, $3^+ \ 1/2$ | $6^+ \ 3/4$, $3^- \ 1/4$ | $1 \ 1/4$, $4 \ 3/4$ | 6^-
3^-: $6^- \ 1/4$, $3^+ \ 3/4$ | $4 \ 3/4$, $-1 \ 1/4$ | $6^+ \ 1/2$, $3^- \ 1/2$ | $1 \ 1/4$, $4 \ 3/4$ | $6^- \ 3/4$, $3^+ \ 1/4$ | 6^+
-1: 3^- | 3^+ | 4 | 6^- | 6^+ | 1
(b) P/222 phases: $\pi_1 = \mathrm{Dic}_3$ (12)
6 4_a 4_b 3 -1
6: $1 \ 1/2$, $3 \ 1/2$ | 4_b | 4_a | $6 \ 1/2$, $-1 \ 1/2$ | 3
4_a: 4_b | $6 \ 2/3$, $-1 \ 1/3$ | $1 \ 1/3$, $3 \ 2/3$ | 4_a | 4_b
4_b: 4_a | $1 \ 1/3$, $3 \ 2/3$ | $6 \ 2/3$, $-1 \ 1/3$ | 4_b | 4_a
3: $6 \ 1/2$, $-1 \ 1/2$ | 4_a | 4_b | $1 \ 1/2$, $3 \ 1/2$ | 6
-1: 3 | 4_b | 4_a | 6 | 1
green: attractive channel; red: repulsive; framed: annihilation possible; shading: number of channels
(c) annihilation probability of a defect with its antidefect (one dot per class)
$P(a\,\bar{a} \to 1)$
1.0
0.5
0.0
C: $\mathbb{Z}_2$
U, D: $\mathbb{Z}_4$
P/222: Dic_3
T: $2T$
O: $2O$